

SNIGDHA ITLA

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Education

National Institute of Technology, Warangal

September 2023 – May 2027

Bachelor of Technology in Electrical Engineering

Warangal, Telangana

Minor in Applied and Computational Mathematics

Relevant Coursework

- Engineering knowledge
- Algorithms Analysis
- Project management
- Systems Programming
- Problem analysis
- Modern tool usage
- Data Structures
- Software Methodology

Projects

Smart Digital Noticeboard System | *Python, Cloud Storage, React.js, Raspberry Pi*

February 2025

- Engineering a smart real-time digital noticeboard powered by Raspberry Pi, updating seamlessly via a Flask API & WebSockets.
- The app allows authorized users to upload notices, which are automatically displayed and removed after expiration.
- Cloud-integrated and secure, using Firebase for storage and JWT authentication for role-based access.
- Optimized power efficiency, reducing energy consumption by 30 percent using cron jobs and Raspberry Pi power management.

Spam Email Detection | *Python, Transformer Models, NLP and Text Processing*

January 2025

- Developed a spam detection system using fine-tuned transformer models (BERT, DistilBERT) for high-accuracy binary classification.
- Boosted accuracy by 12% through advanced preprocessing techniques including tokenization, stopword removal, and embedding extraction.
- Achieved 95% precision on a labeled dataset, validated through F1-score and precision-recall analysis.

Hospital & Property Locator System | *Python, K-Means, Streamlit, Geopy*

November 2024

- Conceptualized and developed an interactive geospatial tool using Streamlit and Geopy to help stakeholders identify optimal property locations based on hospital proximity and population density index.
- Devised a real-time hospital and property locator system with geospatial filtering and interactive maps using Streamlit, improving user accessibility by 40%.
- Designed an interactive, user-friendly map interface displaying hospital counts, property metadata, and population density insights, improving accessibility and decision-making efficiency by 40%.
- Streamlined recommendation accuracy through distance-based filtering (within 5 km radius) using Geopy.

Skills

Languages: Python, C++, HTML/CSS

Frameworks & Tools: Flask, Firebase, Git, GitHub

Core CS: Data Structures & Algorithms, OOP (C++),

Databases: MySQL

Design & UI: Figma, Adobe Photoshop, Canva

Achievements

- Merit Cum Means Scholarship (Top 5 percent) – Awarded for academic excellence at NIT Warangal in the first year.
- Shortlisted for Campus Real-Time Implementation Project – Selected among 13.33 percent (4 out of 30 teams) for an impactful tech solution.
- Led 2 Project Teams driving Software Development and Machine Learning projects at NITW.

Leadership / Extracurricular

- Spearheading a TLR 7 project to enhance campus infrastructure and technology integration.
- Participated in Innothon 2025 by Hitachi Energy, contributing to a project on monitoring and analytics systems for renewable energy & electric mobility and shortlisted among top 10 percent solutions.
- Executive Member of Dance and Dramatics Club, organized and performed in multiple inter-college cultural events, engaging 200+ students.
- Executive Member and Head of Design Team of HAM ARC Technical Club, NIT Warangal, contributing to engineering and technical initiatives.